SurvivalGPU: a GPU-powered library for scalable survival analysis in R and Python

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Pharmacovigilance on the SNDS

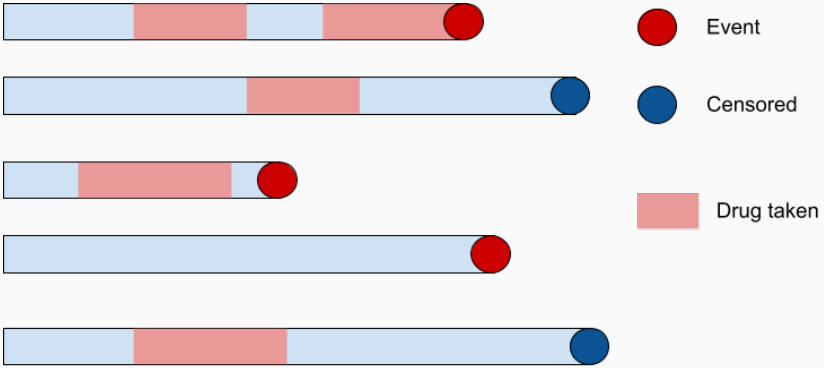
Pharmacovigilance

- Detect adverse drug reactions
- Done by physicians
- Need of a more systematic approach

On the SNDS

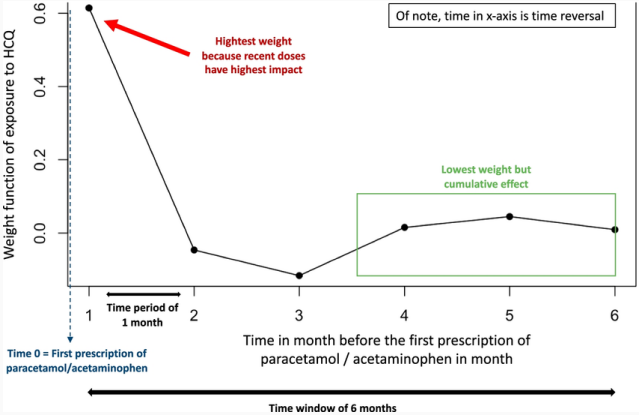
- Data from the social security
- Get all drug prescriptions and hospitalizations
- Almost all the French population

Survival analysis



Longitudinal survival analysis on the SNDS

The Weighted Cumulative Exposure model



The WCE model[1]

[1] Sabatier et al., "A data-driven pipeline to extract potential adverse drug reactions through prescription, procedures and medical diagnoses analysis".

The Weighted Cumulative Exposure model

The WCE formula[1]

$$\mathrm{WCE}(u) = \sum_{t}^{u} w(u-t)X(t)$$

Modelling of w(u - t) by a spline

$$w(u-t) = \sum_{j=1}^{m} \theta_j B_j(u-t)$$

[1] Sylvestre and Abrahamowicz, “Flexible modeling of the cumulative effects of time-dependent exposures on the hazard”.

Le modèle WCE

Integrated in a Cox model

$$h(u \mid \mathbf{X}(u), \mathbf{Z}(u)) = h_0(u) \exp\left[\beta \sum_{t}^{u} w(u-t)X(t) + \sum_{s=1}^{q} \eta_s Z_s(u)\right]$$

By replacing WCE by the spline formula we find a complete cox model

$$h(u \mid \mathbf{X}(u), \mathbf{Z}(u)) = h_0(u) \exp\left[\sum_{j=1}^{m} \gamma_j D_j(u) + \sum_{s=1}^{q} \eta_s Z_s(u)\right]$$

SurvivalGPU

A GPU powered library for scalable survival analysis

- The SNDS has millions of patients
- The WCE model is computationally intensive, especially when using bootstraps
- GPU acceleration is a game changer

In R and Python

- Survival analysis is done by physicians and epidemiologists
- They mostly use R
- We need to adapt to the community

Using the survivalGPU library in R

```r
library(survivalGPU)

model = wceGPU(
        Data = dataset,
        id = "patients",
        Event = "events",
        Start = "start",
        Stop = "stop",
        Expos = "dose",
        Constrained = "Right",
        nknots = 1,
        cutoff = 180,
        nbootstraps= 1000,
        batchsize = 10
    )
```

Creating a R/python interface with reticulate

```
library(reticulate)


use_survivalGPU <- function() {
    python_path <- system.file("python", package = "
survivalGPU")
    survivalgpu <- reticulate::import_from_path("survivalgpu",
 path = python_path)

    return(survivalgpu)
}
```

Calling the Python function from R

```
    wceGPU <- function(data = data, ids = id, covars = py_
covariates, stop = stop, doses = expos, events = event, nknots
 = nknots, constrained = py_constrained, cutoff = cutoff,
bootstrap = nbootstraps, batchsize = batchsize) {

        survivalgpu <- use_survivalGPU()
        wce_r <- survivalgpu$wce_r

        wce_model <- wce_R(parameters)

        [...]

}
```

Adapting the input for Python

```
def wce_R(
    *,data, ids, covars,stop, doses, events, **kwargs,
):

    ids = np.array(data[ids])
    doses = np.array(data[doses])
    times = np.array(data[stop])
    events = np.array(data[events])
    N = len(times)

    res = wce_numpy(ids=ids,covariates=covariates,doses=
doses,events=events,times=times,**kwargs,)

    [...]
```

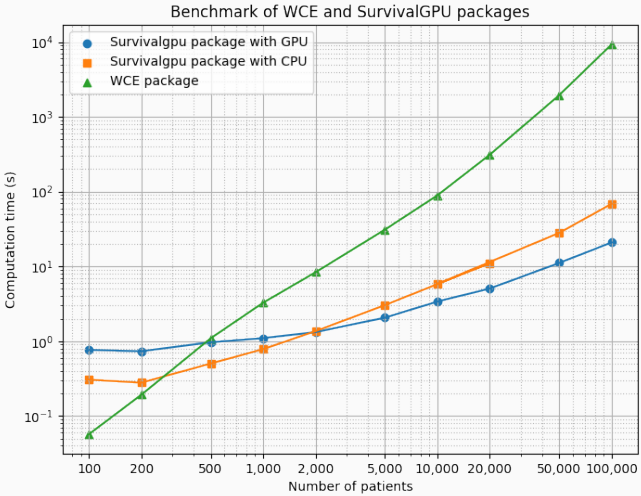
Preparing the input for torch

```
    def wce_numpy(*,ids,covariates,doses,events,times,profile=
None,**kwargs,):

    ids = torch.tensor(ids, dtype=int32, device=device)
    doses = torch.tensor(doses, dtype=float32, device=device)
    times = torch.tensor(times, dtype=int32, device=device)
    events = torch.tensor(events, dtype=int32, device=device)

    result = wce_torch(ids=ids,covariates=covariates,doses=
doses,events=events,times=times,**kwargs,)

    [...]
```

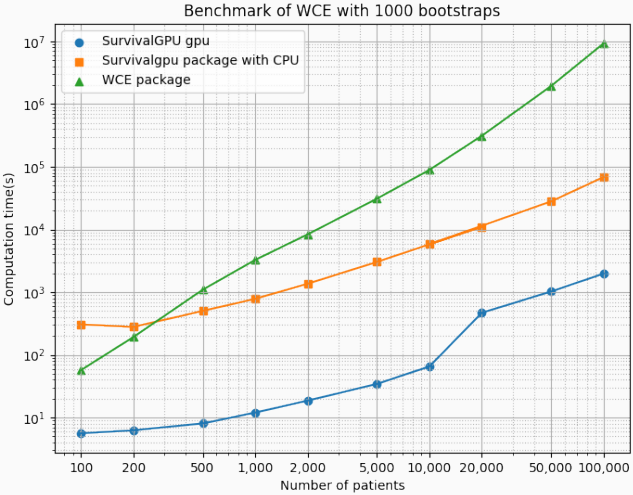
The main function of the WCE model

```
def wce_torch(*,ids,covariates,doses,events,times,cutoff,
constrained=None,nknots=1,order=3,bootstrap=1,batchsize=0,
verbosity=1,):

nknots = int(nknots)
order = int(order)
bootstrap = int(bootstrap)
batchsize = int(batchsize)

[...]
```

Benchmark of WCE



Benchmark of WCE with 1000 bootstraps



The drugdata dataset

Id	Event	Start	Stop	Sex	Age	Dose
1	0	0	1	1	40	0
1	0	1	2	1	40	0
1	0	2	3	1	40	0
1	0	3	4	1	40	0
1	0	4	5	1	40	0
2	0	1	2	0	35	1
2	0	2	3	0	35	1
2	1	3	4	0	35	1

Analysis of the drugdata dataset

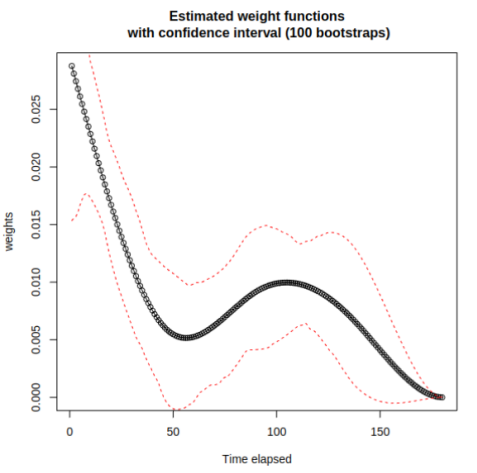
```
model_wce = wceGPU(
            data = drugdata,
            nknots = 3,
            cutoff = 180,
            id="Id",
            event = "Event",
            start = "Start",
            stop = "Stop",
            expos = "dose",
            constrained = "Right",
            covariates = c("age","sex"),
            nbootstraps = 100)

HR_cpu = HR(model_wce, rep(1, 180), rep(0, 180))
plot(model)
```

Covariates results

Variable	coef	CI 2.5%	CI 97.5%	exp(coef)	se(coef)	z	p	Signif.
age	0.012	0.004	0.020	1.012	0.004	3.0	0.003	**
sex	0.707	0.473	0.941	2.028	0.119	5.9	3e-09	***

Results of the analysis of the drugdata dataset

Drug HR : 4.47 (95% CI : 3.32 - 6.78)



Conclusion

- GPU computing is not only for deep learning
- Your skills may be applicable in other fields
- It's important to adapt to the community
- It's easy to create a R wrapper to a python package with reticulate

References

Sabatier, P. et al. “A data-driven pipeline to extract potential adverse drug reactions through prescription, procedures and medical diagnoses analysis: application to a cohort study of 2,010 patients taking hydroxychloroquine with an 11-year follow-up”. en. In: *BMC Medical Research Methodology* 22.1 (Dec. 2022), p. 166. ISSN: 1471-2288. DOI: 10.1186/s12874-022-01628-3. URL: https://bmcmedresmethodol.biomedcentral.com/articles/10.1186/s12874-022-01628-3 (visited on 12/15/2023).

References ii

Sylvestre, Marie-Pierre and Michal Abrahamowicz. “Flexible modeling of the cumulative effects of time-dependent exposures on the hazard”. en. In: *Statistics in Medicine* 28.27 (Nov. 2009), pp. 3437–3453. ISSN: 02776715, 10970258. DOI: 10.1002/sim.3701. URL: https://onlinelibrary.wiley.com/doi/10.1002/sim.3701 (visited on 10/25/2023).